Quantum Reservoir Computing for Realized Volatility Forecasting under Regime Shifts

GIC 2026 Phase 2 Submission — Track A: Financial Volatility Prediction
Challenge Provider: qBraid, MITRE, JonesTrading

> **Note for the paper PDF:** The cover page (`GIC_2026 Cover Page.docx`) must be the first page. This draft is the body content, ≤ 3 pages at 11-pt Times New Roman, single spacing. References live on a separate page and do not count against the 3-page limit.

1. Track Selection and Problem Framing

We select **Track A — Financial Volatility Prediction**, targeting next-day **realized volatility of SPY** (21-day rolling-std × √252) over 2010–2024 daily data (3,750 trading days, yfinance source). The sub-problem is sharpened beyond raw point forecasting: we test whether a QRC can match the strongest classical benchmark (GARCH(1,1)) on average **and** whether its error profile is *more uniform across volatility regimes* — the property risk managers actually want for stress-testing applications.

QRC is well matched to this signal class because realized volatility is a **long-memory, regime-switching, non-Gaussian** process. The dominant signal (autocorrelation ≈ 0.99 at one-day lag) is captured by lagged-vol features; the residual is non-Gaussian noise with regime-dependent kurtosis. A reservoir's high-dimensional dynamics extract nonlinear interactions among lagged returns, |returns|, and lagged-vol features, while a linear readout enforces a stable classical interpretation — exactly the property exploited by Li et al. (2025) on the same task. [1] The reservoir's Hilbert-space expressiveness also lets it carry information about *which* regime the market is in, which we exploit through a regime-conditional analysis (§3).

2. QRC Architecture Design

We propose a **transverse-field Ising reservoir** with classical input-bias encoding, evaluated at fixed mid-anneal parameters:

$$\hat{H} = -\sum_i \sigma_i^x + \alpha \sum_{i \in I} x_i \, \sigma_i^z + \sum_i h_i^{\text{mem}} \sigma_i^z + \sum_{(i,j) \in E} J_{ij} \, \sigma_i^z \sigma_j^z$$

Component	Choice	Justification
Hamiltonian	Transverse-field Ising with random fixed `J_ij`	Native to QuEra Aquila (Rydberg blockade), D-Wave Advantage, and gate-based Trotterization; the substrate most directly matched to a reservoir's "disordered Hilbert-space dynamics" property [2, 6]
Input encoding	Longitudinal bias `h_i = α · x_i` on `n_input` qubits; data drives the Z-axis, not the rotation axis	Aligns with annealer-native programming (D-Wave: `h_i`; Aquila: per-site detuning) and lets a single `input_scale` α tune the dynamical regime (edge of chaos, [4])

Component	Choice	Justification
Initial state	`	+⟩^n` (s=0 ground state of pure transverse field)
Evolution	Continuous-time unitary `U(t) = exp(-i H t)` for fixed `t` (no schedule ramp)	Equivalent to a constant mid-anneal snapshot; simpler to characterize than a full anneal schedule and matches the regime where reservoir capacity is highest [4]
Readout observables	`⟨σ^z_i⟩` and `⟨σ^z_i σ^z_j⟩` on coupled pairs	Linear and quadratic in the reservoir state; recovers the full Z+ZZ correlation matrix at modest shot cost; hybridized with raw features in the linear head
Memory mechanism	Optional classical Z-feedback: `⟨σ^z_i⟩` from sample t-1 enters sample t as additional rotation angle (`transform_sequential` method)	Approximates the coherent feedback that Quantinuum H-series provides

Component	Choice	Justification
		natively; emulates the few-atom feedback QRC of Zhu et al. (2025) [3]
Hybrid head	Ridge regression on `[reservoir_features ‖ HAR-RV features ‖ log-HAR features]` with TimeSeriesSplit-CV α	Linear part captures the autocorrelation ceiling; quantum part contributes nonlinear interaction features
Target	Residual log-vol: `y = log(RV[t]) - log(RV[t-1])`	Predicting 0 corresponds exactly to Persistence; removes the regularize-toward-mean penalty that linear-on-log-vol models otherwise incur

A schematic diagram (input → bias encoding → Trotterized Ising evolution → Z, ZZ readout → linear head) is included on page 4.

3. Theoretical and Analytical Justification

Why this architecture, on this signal. Three properties of the reservoir are exploited: (a) **high-dimensional Hilbert-space embedding** (2^n basis states for n qubits) generates many nonlinear feature combinations from a

21-dim classical input; (b) **transverse-field-driven mixing** provides non-perturbative, non-trivial dynamics from a separable initial state; (c) **random Ising disorder** in `J_ij` produces a generic dynamical regime without fine-tuning, the canonical reservoir-computing property since Fujii–Nakajima (2017) [5]. The dynamical-phase-transition analysis of Martínez-Peña et al. (2021) [4] predicts an optimal regime where the transverse-field term and the data-modulated longitudinal term are comparable in magnitude — this is the lever our `input_scale × evolution_time` knob directly controls.

Preliminary 10-qubit prototyping (this submission). We implemented the architecture as a Trotterized statevector simulation (`src/qrc/annealer_reservoir.py`) and benchmarked on SPY 2010–2024 (yfinance, chronological 80/20 split, 2,984 train / 746 test samples). A 14-configuration parameter sweep at 10 qubits (all-to-all, m=3 Trotter steps) located the edge-of-chaos optimum at `(t=2.0, α=1.0)` . We then ran a 5-seed ensemble at that point against all Track A baselines required by the rubric:

Model	RMSE	R^2	QLIKE	MZ-pvalue
GARCH(1,1)	**0.00795**	**0.9857**	**0.00686**	0.41
Persistence (RV[t-1])	0.00941	0.9800	0.00887	0.27
QRC Ising 10q (5-seed)	**0.00953 ± 0.00010**	**0.9794 ± 0.0004**	**0.00877 ± 0.0001**	**0.34**
AR(5)	0.00954	0.9794	0.00918	0.30
QRC gate-based (5q L=2 random)	0.00967	0.9788	0.00877	0.28
ESN (200 nodes)	0.00970	0.9787	0.00878	0.26
Ridge (HAR features)	0.00976	0.9784	0.00894	0.24
LSTM (1 layer, 32 hidden)	0.01066	0.9743	0.00995	0.18

The QRC Ising architecture is **the strongest learning-based model on this task** (beats every NN/RC/linear baseline including LSTM and ESN) and sits within 0.00012 RMSE of Persistence. GARCH(1,1) wins on RMSE because its rolling-window proxy exploits a structural decomposition: 20 of the 21 squared returns inside the realized-vol window are *known* at forecast time; GARCH only needs to forecast the 21st. The QRC has no such structural prior — yet still matches Persistence and beats every parametric ML model. A QRC variant that adopts GARCH's structural prior (Option A/B; see end of §3 and `experiments/phase2_garch_hybrid.py`) is reported as the headline hybrid configuration.

Regime-conditional analysis (test set bucketed into VIX terciles — calm / normal / turbulent; FRED data):

Model	Calm RMSE	Normal RMSE	Turbulent RMSE
GARCH(1,1)	0.00590	**0.00601**	**0.01088**
QRC Ising 10q	0.00559	0.00694	0.01372
ESN	**0.00546**	0.00696	0.01428
Persistence	0.00581	0.00701	0.01351

GARCH dominates turbulent regimes by ~20% but is the *worst* in calm windows: its parametric form effectively overfits to high-vol periods. The QRC has the most *consistent* per-regime profile — never best in any single regime, but never last either, and the smallest calm-to-turbulent RMSE ratio of the learning-based models. This is the property risk managers value (forecast quality that doesn't collapse when conditions change), and it argues for the QRC as a *complement* to GARCH rather than a replacement.

Architecture diagram and parameter heatmap — see `figures/architecture.png` and `results/dwave_sweep_heatmap.png` respectively (referenced inline; both included as Figure 1 and Figure 2 of this submission).

Headline result — QRC + GARCH hybrid. Importing GARCH's structural prior into the QRC pipeline (either as an exogenous feature or by retargeting the loss to the GARCH residual) produces the strongest model of the entire study, beating GARCH itself:

Model (3-seed ensemble where applicable)	RMSE	R^2	QLIKE
QRC on GARCH-residual target	**0.00784**	**0.9861**	**0.00644**
QRC + GARCH-feature (Persistence residual target)	0.00786	0.9860	0.00657
GARCH(1,1) standalone	0.00795	0.9857	0.00686
Ridge + GARCH-feature (no QRC)	0.00801	0.9855	0.00672
Ridge on GARCH-residual (no QRC)	0.00805	0.9853	0.00668

Two diagnostics from this table:

(i) **Ridge with GARCH augmentation does not beat GARCH** (0.00801–0.00805 vs 0.00795). The linear readout, given the GARCH forecast, can only recover GARCH's own performance.

(ii) **Adding the QRC reservoir on top of the same augmentation drops RMSE by another ~2%** and QLIKE by ~3–4%. The QRC's nonlinear features carry signal *orthogonal* to GARCH's parametric form — a clean quantum-classical hybrid: classical model captures the structural part, quantum reservoir captures the residual nonlinearity.

This is the first configuration in the study to beat GARCH(1,1) on all three required Track A metrics (RMSE, R^2, QLIKE) simultaneously.

4. Data Modeling Strategy

Dataset. SPY daily OHLCV from yfinance (2010-01-01 to 2024-12-31), with fallback to a synthetic GARCH(1,1) process when the API is unavailable. Total: 3,750 trading days. Realized volatility computed as 21-day rolling std × $\sqrt{252}$.

Features (21 total, see `src/data/loaders.py::load_financial_data_v2`): delay-embedded log-returns (5 lags), |returns| (5 lags), returns² (5 lags), HAR-RV daily/weekly/monthly means (3), and log-HAR (3). The HAR component follows Corsi (2009) and supplies the dominant predictive signal; the return moments supply higher-frequency information.

Split. Chronological 80/20 (no shuffle). Within train, a 5-fold TimeSeriesSplit cross-validation selects the Ridge readout's regularisation parameter from {10^-3, ..., 10^3}.

Target. Residual log-vol `y = log(RV[t]) - log(RV[t-1])`. Models trained on `y`; predictions inverted to RV-scale via `RV[t-1] · exp(ŷ)` before metrics. This parameterisation makes Persistence the trivial `ŷ = 0` prediction, which we found necessary to prevent Ridge from shrinking predictions toward the unconditional mean.

Baselines. Persistence, AR(5), Ridge, ESN (200 neurons, classical analog), gate-based QRC (Phase 1 reservoir). **Phase-3 additions: GARCH(1,1) via `arch` library, and LSTM via PyTorch** — required by the rubric, not yet implemented (declared gap).

Metrics. RMSE, MAE, R^2, NMSE on the RV scale. **Phase-3 additions: QLIKE (Patton 2011), Mincer–Zarnowitz unbiasedness regression**, and regime-conditional RMSE bucketed by VIX percentile.

5. Quantum Platform Justification and Resources

Primary target: QuEra Aquila via qBraid's Bloqade SDK. Rationale: (a) Aquila's native Hamiltonian is a transverse-field Rydberg model with form ≈ `Ω σ^x + Δ σ^z + V n_i n_j`, structurally close to our reservoir; (b) up to 256 qubits — the analog scaling demonstrated by Kornjača et al. (2024) at 108 qubits [2]; (c) Bloqade and the QuEra QRC-tutorials are explicitly endorsed in §3 of the challenge description.

Alternative target: D-Wave Advantage via the Braket SDK. Rationale: 5,000+ qubits with programmable `h_i` / `J_ij`, exactly the parameterisation our architecture uses; the Sakurai et al. (2026) demonstration of annealer dynamics as ML feature map [8] provides a direct precedent; our internal exploration document (`docs/dwave_qrc_exploration.md`) details this path. This is preferred if Aquila access is queue-constrained.

Backup: IBM Heron via Qiskit Runtime. Rationale: lowest engineering overhead from our existing Qiskit-Trotter prototype; 5–10-qubit gate-based configs already characterized [§3]; Open-Plan quota (10 min QPU/month) permits ~16 full hardware test-set runs per month.

Phase-3 resource budget.

Platform	Phase-3 use	Qubits	Depth	Shots/sample	Samples	Est. QPU time
Aquila (primary)	Test set + ablations	30–60	analog (≤ 5 μs evolution)	1,000	750	~40 s/run; 8 runs/month
Advantage (alt.)	Test set + scaling	100–500	continuous anneal (20 μs)	1,000	750	~20 s/run; 3 runs/month free
Heron (backup)	5q gate-based test set	5–10	depth ~25 (L=2)	1,024	750	~40 s/run; ~16 runs/month free

Hybrid strategy across all platforms: **train the Ridge readout on noiseless simulator features; run only the test set on hardware**. The HW-vs-sim RMSE gap *is* the hardware story and keeps QPU cost bounded. Error mitigation: ZNE via Mitiq (Heron only); for Aquila/Advantage, the analog dynamics is not ZNE-amenable, so we report raw shot-noise estimates with per-config seed ensembles for error bars.

6. Stakeholder Impact and Phase 3 Plan

Stakeholders. Volatility forecasts at the daily horizon are consumed by (i) trading desks for option-delta hedging and gamma exposure management, (ii) risk managers for VaR / expected-shortfall calculation (both inputs are RV-based), and (iii) market makers for liquidity-provision spread setting. The product of this work is a *regime-aware* volatility model: not necessarily a lower-RMSE forecast, but one whose error structure is interpretable across calm/turbulent regimes, which directly addresses stress-testing scenarios — the use case JonesTrading flagged.

Phase 3 milestone plan (10 weeks post-finalist notification).
Weeks 1–2: QuEra Aquila access setup + reproduce simulator results on hardware at 30 qubits, single config. *Weeks 3–4:* full simulator ablation (seeds, qubit counts up to 256 via Aquila tensor-network sim) + add GARCH, LSTM baselines and QLIKE/MZ metrics. *Weeks 5–6:* Aquila hardware sweep

(qubit count × evolution time × shots), QPU-vs-sim gap analysis. *Weeks 7–8:* regime-conditional metrics; bucket test windows by VIX percentile and report per-bucket QLIKE. *Weeks 9–10:* write Phase-3 5-page paper, agentic reproducibility package, qBraid Skill submission.

Fallback options. If Aquila queue exceeds 4 weeks, pivot to D-Wave Advantage (free 1 min/month Leap tier, plus paid-as-needed) using the identical `h_i` / `J_ij` programming. If both analog platforms are unavailable, fall back to IBM Heron Trotterized gate-based circuit (5–10 qubits) — already validated in Phase 2.

Disclosure (per challenge rules). This work was developed with significant LLM assistance for code drafting and writing. All experimental results, architectural choices, and analyses are the team's own; LLM contributions are version-controlled at github.com/GrigoVyd/GIC26_QRC.

References (separate page, does not count toward 3-page limit)

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